

Troubleshooting

Problems in the order people actually hit them.

Quick triage

Symptom	Most likely cause	Jump to
Agent doesn't move at all	No navigation manager in the level	§1
"No path" through a gap you can see	<code>Clearance Padding</code> sealed it	§3
Agent flies through walls	Geometry not voxelized	§4
Moving obstacles ignored	Missing Dynamic Obstacle component	§5
Can't find the demo maps	<code>Show Plugin Content</code> is off	§7
Frame rate drops	Voxel resolution too fine	§6
Long paths fail, short ones work	Infinite-world streaming	§3

The two diagnostics that answer almost everything

1. Turn on `Draw Debug Volumes` (navigation manager → 5. *Debug*).

This draws the octree. It tells you what the pathfinder actually sees, which is frequently not what you assume:

- No small voxels around a doorway → the pathfinder can't route through it either.
- No voxels at all in a region → it's outside `World Extent`, or the chunk hasn't generated.
- Solid voxels where there's visibly open air → `Clearance Padding` is too large.

2. Check the **Output Log** (Window → Output Log) and filter for `Lcm`. The plugin logs the specific reason a query failed.

1. Nothing moves

Work in this order.

Is there a navigation manager?

Output Log shows `LCMBTTask_FlyTo: No SVO Manager found in the level!` → drag an `LcmNavigationManagerSVO` into the level. One per level, required.

Did the AI possess the pawn?

- Pawn → Class Defaults → `Auto Possess AI` = `Placed in World or Spawned`
- Pawn → Class Defaults → `AI Controller Class` is set
- The controller actually runs the tree (`Run Behavior Tree` , or a `StateTreeAIComponent` with a State Tree assigned)

Can the pawn move at all?

It needs a movement component that supports flight. `Floating Pawn Movement` is the simple choice. Check its `Max Speed` isn't 0.

Is the goal set?

Print the blackboard key (or StateTree parameter). A goal of `(0,0,0)` means it was never set, or was set after the tree started.

2. Agent moves, but badly

Symptom	Cause	Fix
Jitters, vibrates in place	Physics fighting navigation	Set mesh collision to <i>No Collision</i> to confirm, then re-add carefully
Stops short of the goal	<code>Acceptance Radius</code> too large	Lower it, but not below one voxel
Overshoots and circles	Speed too high for turn rate	Lower <code>Flight Speed</code> or raise <code>Deceleration Distance</code>
Robotic square turns	Grid-locked solver, no smoothing	Use <i>Lazy Theta*</i> and <i>Curved Spline</i> smoothing
Sinks into the floor	Insufficient clearance below	Raise <code>Clearance Padding</code> slightly, or <code>Min Voxel Size</code> finer
Agents pile up on each other	No separation	Raise <code>Separation Radius</code> / <code>Separation Weight</code> , or use <i>Predictive (ORCA)</i> avoidance

3. "No path found"

First: is the goal reachable at all?

- Inside the navigation volume? Outside `World Extent` there is no data. Enlarge it, or move the goal in.
- Inside geometry? A goal (or start) buried in a wall has no free voxel. Move it into open air.
- Genuinely enclosed? If the destination is sealed off, failure is correct.

The classic: padding sealed the opening

This is the most common cause of a mystifying "no path".

If `Clearance Padding` is close to `Min Voxel Size`, obstacle inflation fills narrow openings completely. A 2 m doorway with 100 cm voxels and 100 cm padding can vanish entirely.

Fix: lower `Clearance Padding`, or lower `Min Voxel Size` so the opening survives the inflation. Confirm with `Draw Debug Volumes`: you should see free voxels *through* the opening.

Short paths work, long ones fail

Almost always an **infinite world** issue: the route crosses chunks that haven't generated yet.

- Ensure `Persistent Skeleton` is on (it is by default). It's what allows long-range planning before the detail exists.
- For routing to be complete on the very first frame, bake a skeleton (`Bake Skeleton (Editor)`) and tick `Use Baked Skeleton`.
- Tick `Assist Infinite Streaming` on the Fly To task for agents that must traverse the frontier.
- Leave `Chunk Size` at 8000 unless you have a specific reason and will re-test long routes.

Everything fails immediately

The volume never built. Check `Auto Build On Play` is on, or that you call `BuildSVO`. With `Draw Debug Volumes` on, you should see the octree at `BeginPlay`.

4. Agents ignore geometry

If agents fly through a wall, the wall isn't in the octree.

1. Does the mesh have collision? Voxelization uses collision, not visual geometry. *No Collision* = invisible to navigation.
2. Is its channel in `Obstacle Query Channels`? (3. Collision)
3. Is `Allowed Obstacle Classes` filtering it out? Empty = allow everything, which is what you normally want.

4. **Was it added after the build?** Static geometry is baked at build time. Something spawned later needs either a rebuild or a Dynamic Obstacle component.
5. **Is it too thin?** Geometry thinner than a voxel can be missed. Lower `Min Voxel Size`, or thicken the collision.

5. Dynamic obstacles do nothing

1. Does the actor have an `LCM Nav3D Dynamic Obstacle` component? Moving an actor without one changes nothing as far as navigation is concerned, and produces no error.
2. Does its mesh have collision enabled?
3. Is `Update Distance Threshold` too high? It won't restamp until the actor moves that far (default 50 cm).
4. Reaction too slow? Lower `Finite Obstacle Update Interval` (default 0.1 s).
5. Using `Tracked Components` ? If the list is non-empty, only those components are tracked, so make sure the blocking one is in it.

See [Tutorial 05](#).

6. Performance

Build is slow / hitches when streaming

- Raise `Min Voxel Size` (the biggest lever by far).
- Keep `Use GPU Voxelization` on.
- Lower `Max Generations Per Frame` to spread work (world fills in more slowly, but smoothly).

Frame rate drops with many agents

- Lower `Max Path Requests Per Frame` to cap per-frame planning.
- Use cheaper avoidance: *Reactive* rather than *Predictive*.
- For agents sharing a goal, enable `Use Flow Fields`.
- Past a few hundred agents, move to `Mass Entity`. Actor ticks, not pathfinding, are the ceiling.

Memory too high

- Raise `Min Voxel Size` (memory scales steeply with resolution).
- Shrink `World Extent` to just your play space.
- In infinite worlds, keep `Chunk Size` at the default.

Debug drawing is itself slow

Expected: it's a debug tool. Raise `Debug Redraw Coalesce Seconds`, or turn it off.

7. Editor and install

Problem	Fix
Can't find demo maps	Content Browser → Settings → Show Plugin Content
"Modules are missing or built with a different engine version"	Rebuild; if it fails, your project is Blueprint-only. See Tutorial 01
Plugin absent from the Plugins window	Wrong folder layout: must be <code>YourProject/Plugins/LCMNav3D/LCMNav3D.uplugin</code>
Fails to load, <code>GetLastError 126</code>	A Mass plugin dependency got disabled; re-enable <code>MassEntity</code> , <code>MassGameplay</code> , <code>MassAI</code>
Compile errors about StateTree or Mass types	Same cause: an engine plugin dependency is off
EQS preview shows nothing in-editor	Call <code>Build SVO In Editor</code> on the manager; EQS previews need navigation data without playing

8. Determinism and multiplayer

If you need identical results across runs (replays, lockstep, automated tests):

- Set compute to **CPU**, not `Auto` or `GPU`.
- Turn `Use GPU Voxelization` off.

GPU execution order isn't guaranteed run-to-run. The CPU path is deterministic by design.

Still stuck?

Gather these before asking for help, since they identify most problems immediately:

1. `Draw Debug Volumes` screenshot of the area that misbehaves.
2. Output Log filtered to `Lcm`.
3. Manager settings: `Infinite World`, `World Extent` / `Chunk Size`, `Min Voxel Size`, `Clearance` `Padding`.
4. Which path you're using: Behavior Tree, StateTree, or Mass.
5. Does the equivalent demo map work? If yes it's a setup issue in your level; if no it's an install issue.

Chapter 14 of the LCM Nav3D User Manual.

The complete manual is `LCM_Nav3D_Manual.pdf`, in this folder alongside every other document in the set. Start from `README.pdf`.

LCM Nav3D: True Volumetric 3D Navigation for Unreal Engine. Version 2.05 · Unreal Engine 5.8.

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